UNIVERSITY OF CAMBRIDGE

Cosmological and Astrophysical Updates From BayeSN

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erc European Research Council Established by the European Commission

Calculating Distance

$$\mu^s = m_B^s - M + \alpha x_1^s - \beta c^s$$

The Impact of Dust

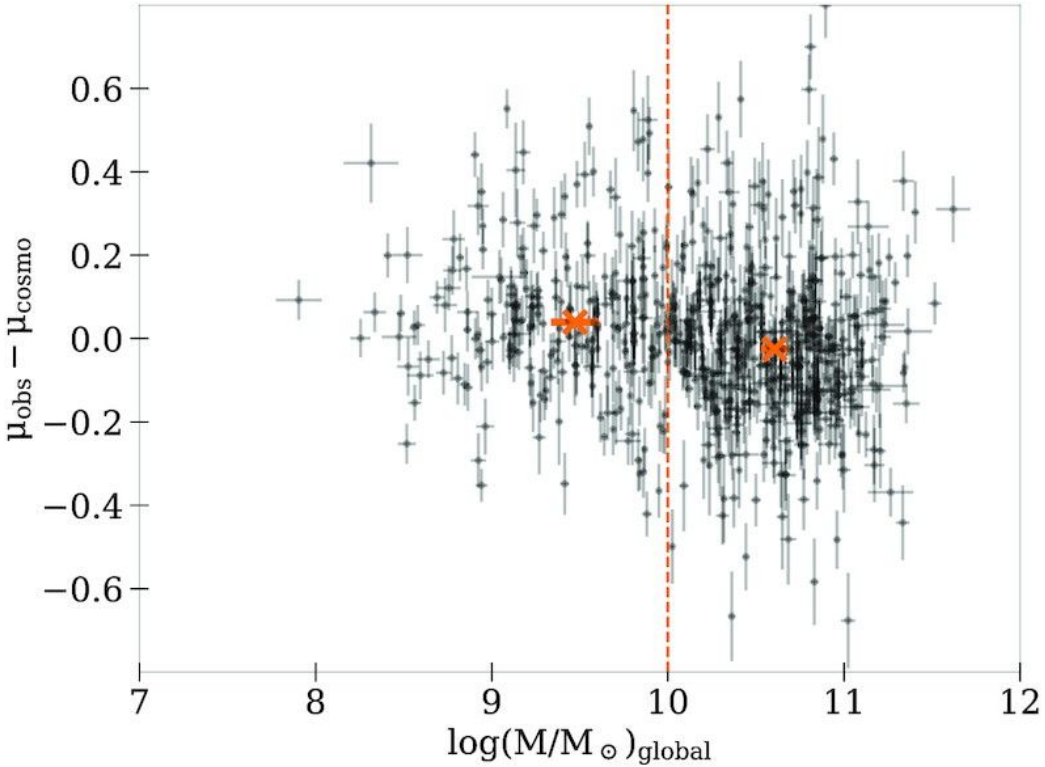
$$\mu^s = m_B^s - M + \alpha x_1^s - \beta c^s$$

$$c^s = c_{\rm int}^s + E(B-V)^s$$

- Colour - one parameter for two separate effects
- Correctly handling dust is key to uncover intrinsic SN properties and accurately estimate distances

The Mass Step

- SN Ia properties depend on environment
- Just different dust properties?
  - SN populations identical, just appear different?
- Or intrinsic differences?
  - Interesting physical implications, different progenitor channels?



Kelsey+2023

Recipe for a Type Ia Supernova Model

Type Ia SN SED

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Recipe for a Type Ia Supernova Model

Type Ia SN SED

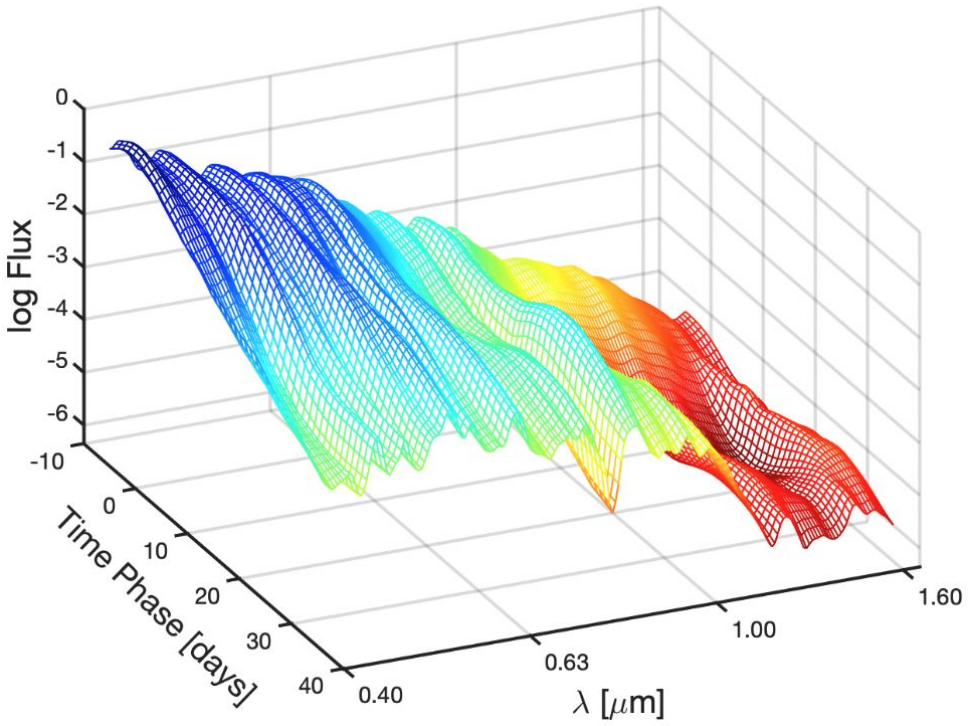
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Mean Intrinsic SED

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Recipe for a Type Ia Supernova Model

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Mean Intrinsic SED

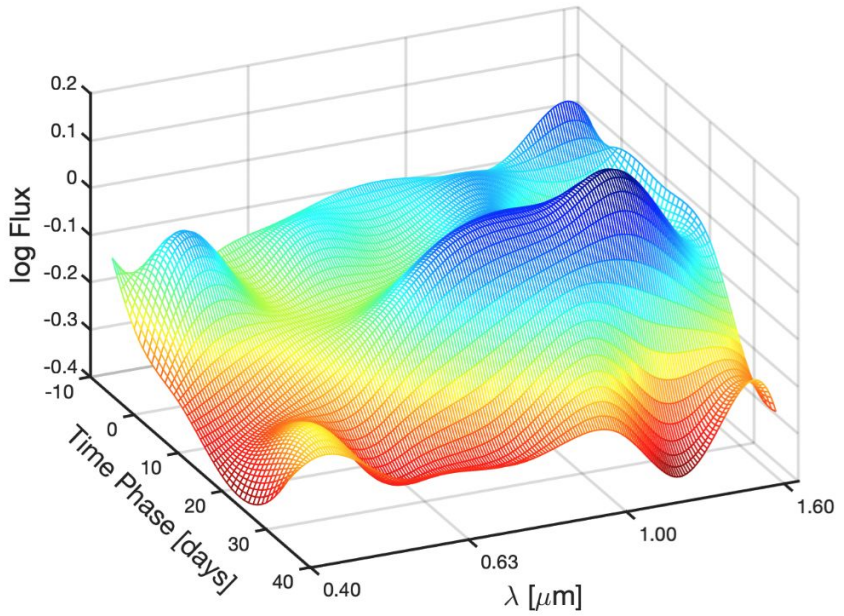
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Effect of Stretch

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$\theta_1 \times$



Recipe for a Type Ia Supernova Model

Type Ia SN SED

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Mean Intrinsic SED

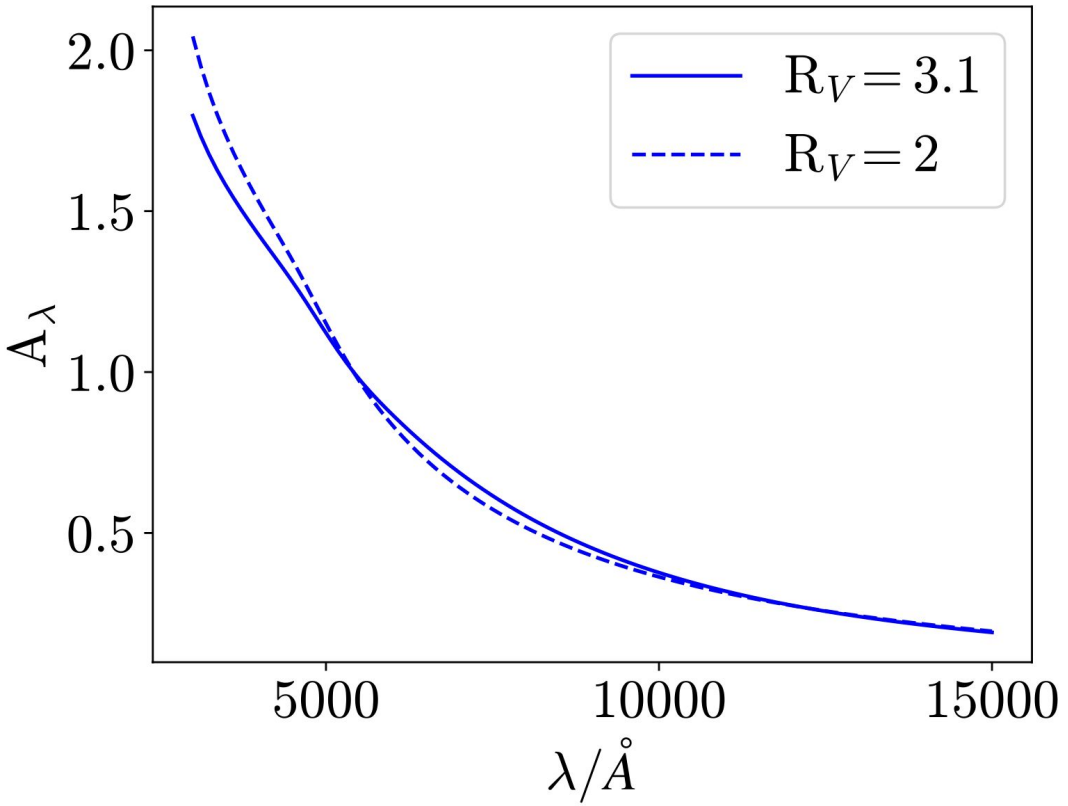
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Effect of Stretch

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Dust Extinction

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Recipe for a Type Ia Supernova Model

Type Ia SN SED

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Mean Intrinsic SED

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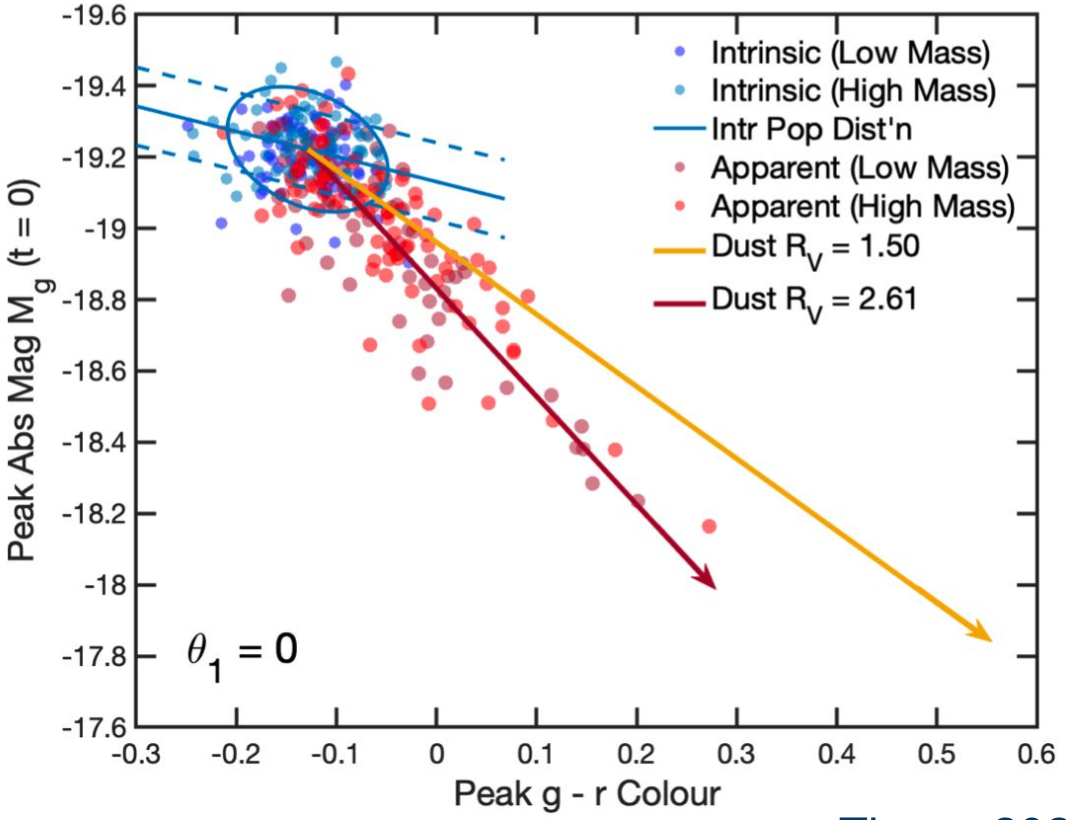
Effect of Stretch

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Dust Extinction

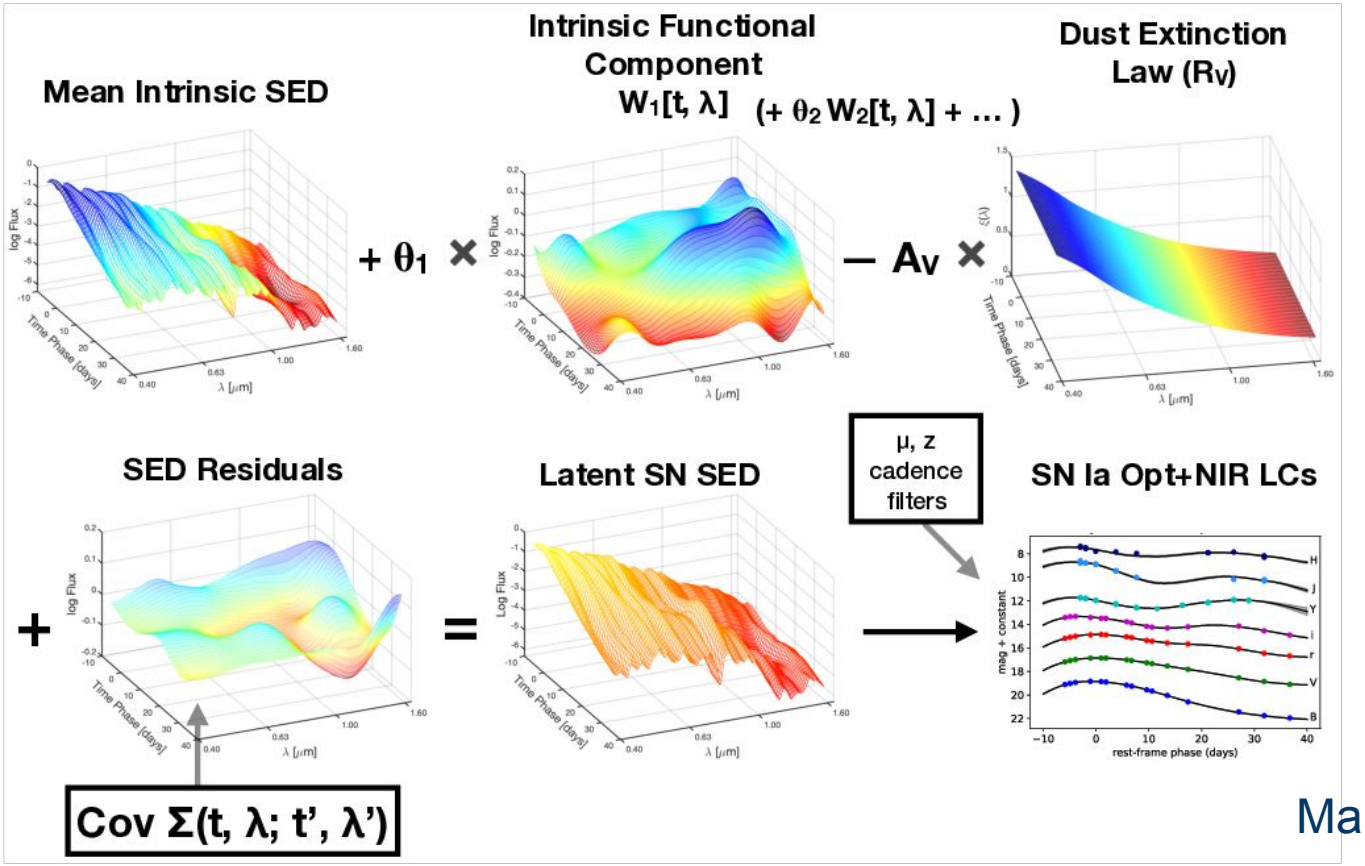
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Intrinsic Colour



Thorp+2021

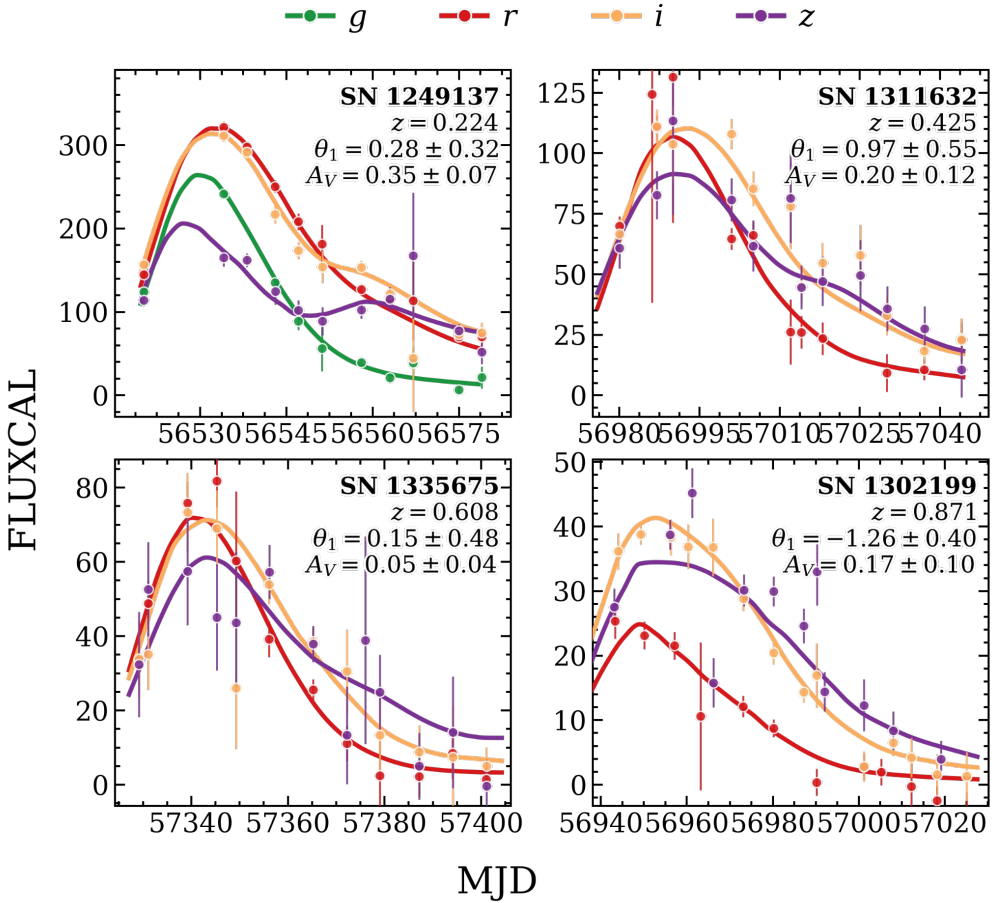
The BayeSN Model



Mandel+2020

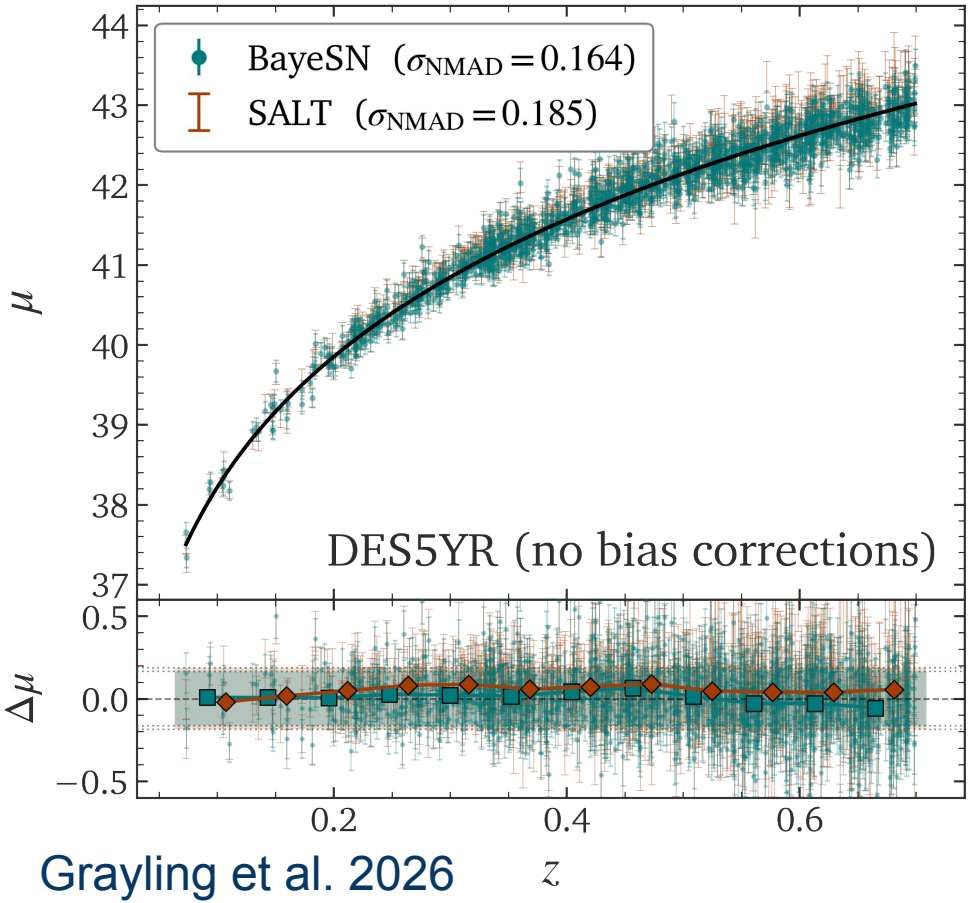
Advantages of BayeSN

- Hierarchical Bayesian Model
  - Separate treatment of dust and intrinsic variation
- Full SED model
  - Uses all bands, not single colour/magnitude pairs
- Model populations, fit light curves
  - Infer dust/intrinsic distributions
  - Fit for SN distances
- Optical-NIR model



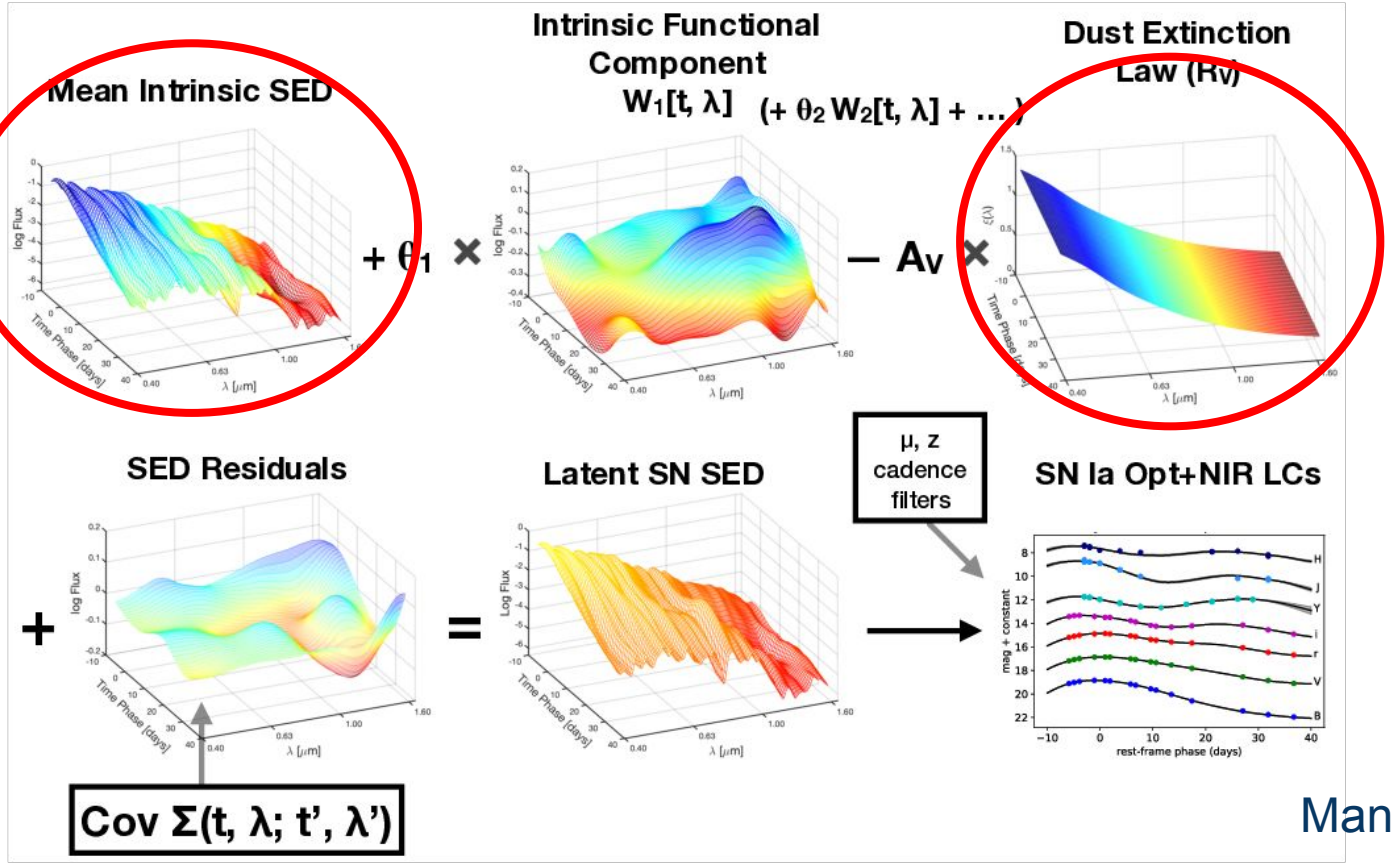
Light Curve Fitting with BayeSN

- Newly released G26 BayeSN model, 12% reduced scatter on DES5YR compared with SALT
- First joint modelling of SN SED with cross-calibration systematics
  - Dovekie as a prior
- Publicly available now!



Grayling et al. 2026

Investigating the Mass Step



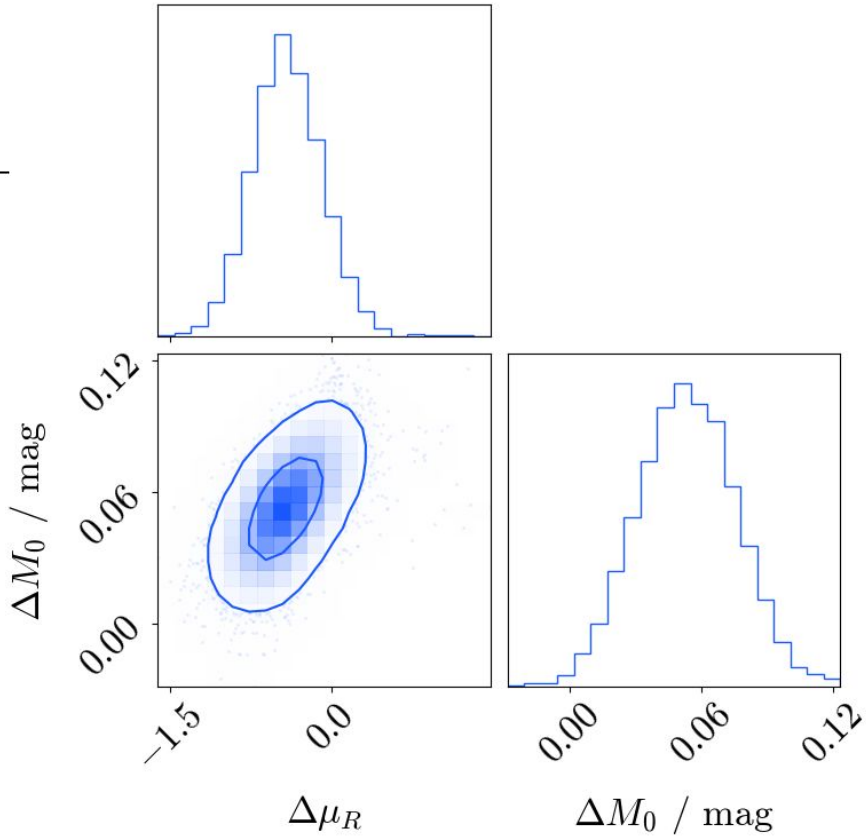
Mandel+2020

Cause of the Mass Step

Sample	Intrinsic Mass Step / mag
Combined*	0.049 ± 0.019
DES5YR	0.053 ± 0.022

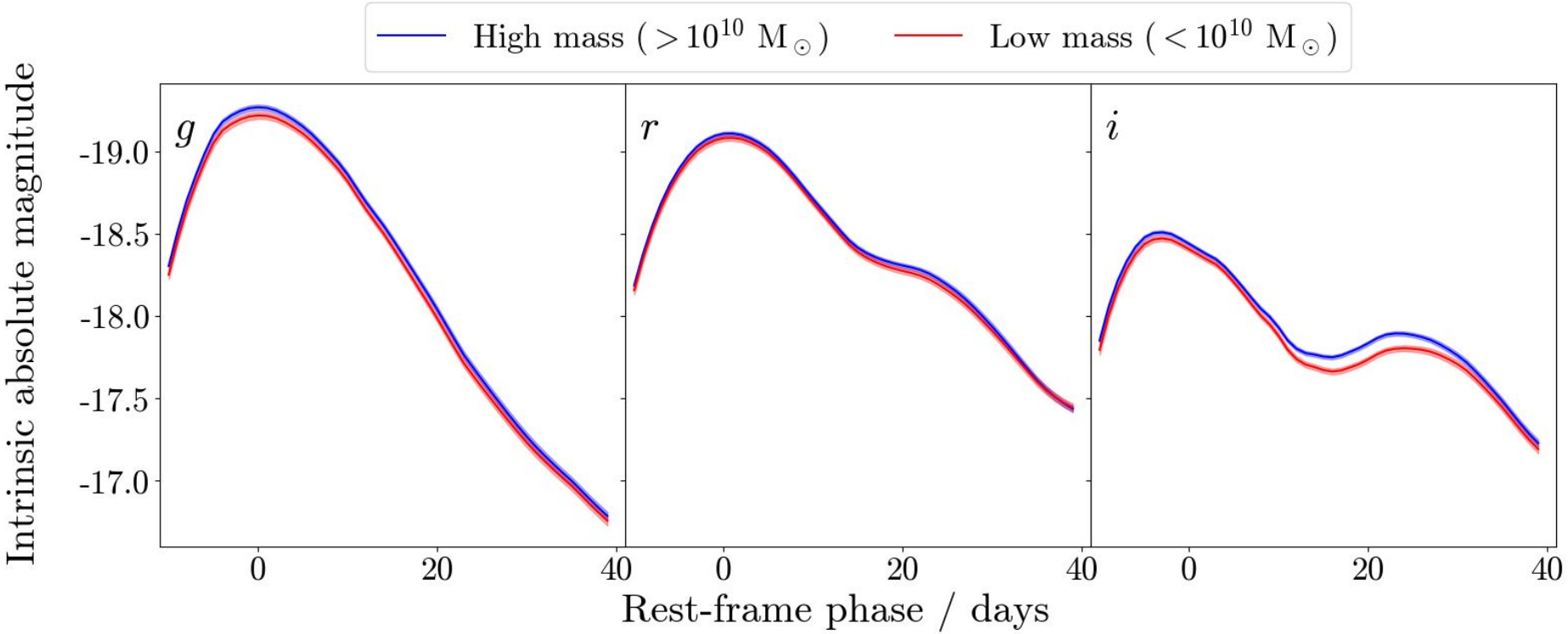
*Combined = DES3YR+Foundation+PS1MD

Dust may play a role, but BayeSN supports an intrinsic contribution to the mass step



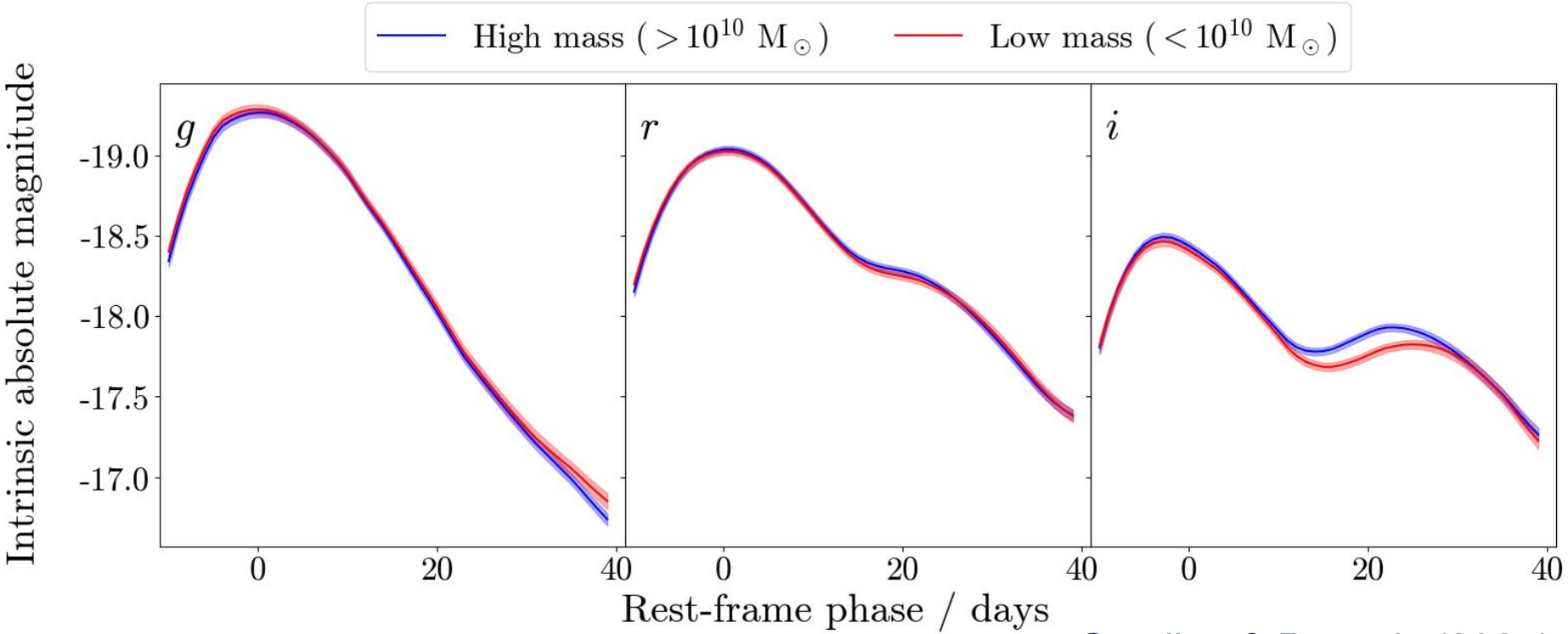
Grayling & Popovic (2025)

DES3YR + Foundation + PS1MD



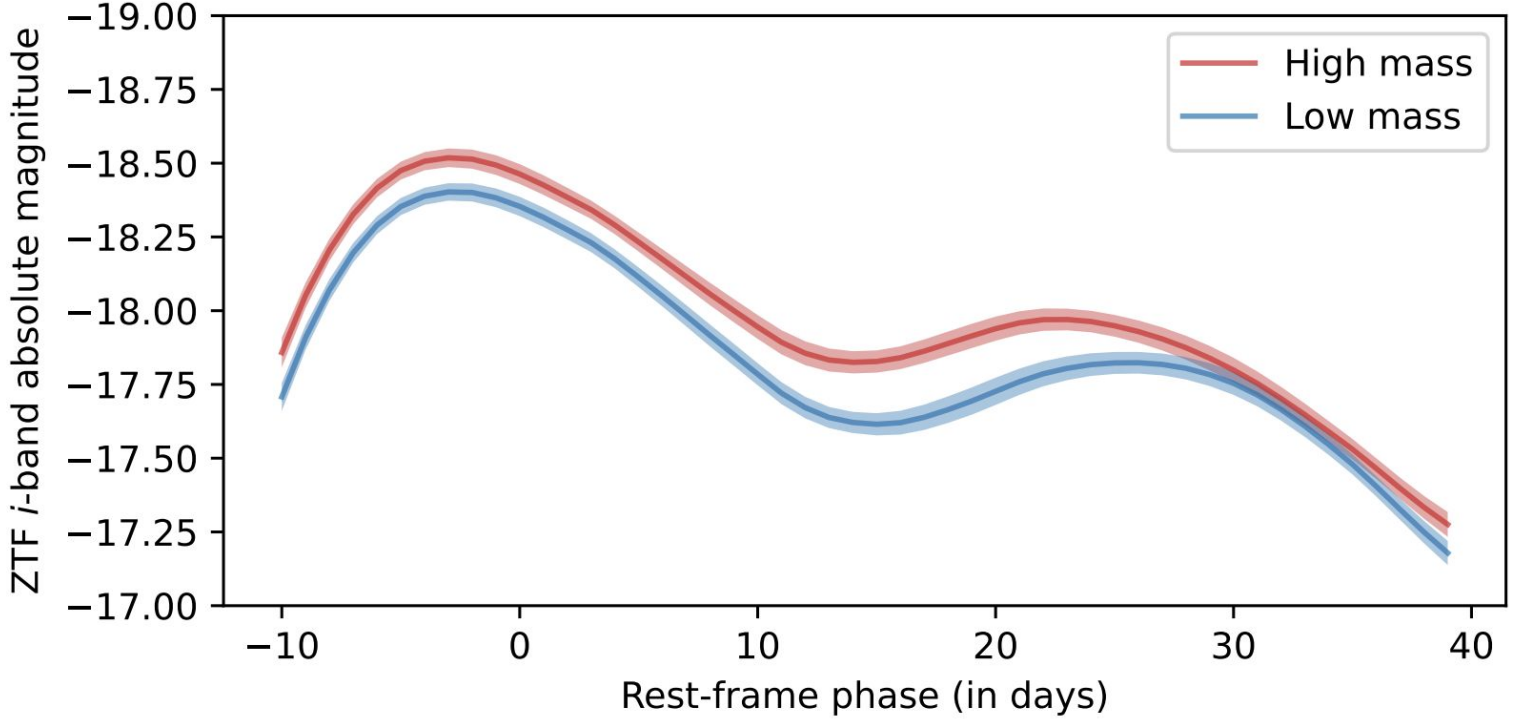
Grayling+2024

DES5YR



Grayling & Popovic (2025)

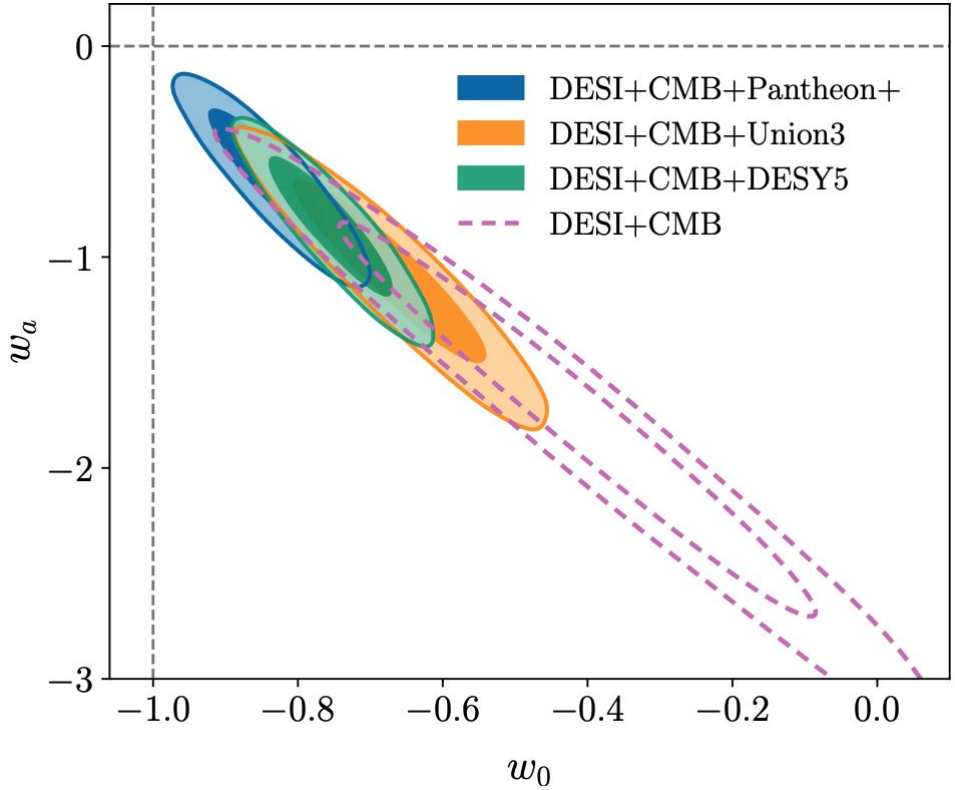
ZTF



Ginolin+2026

What Does This Tell Us?

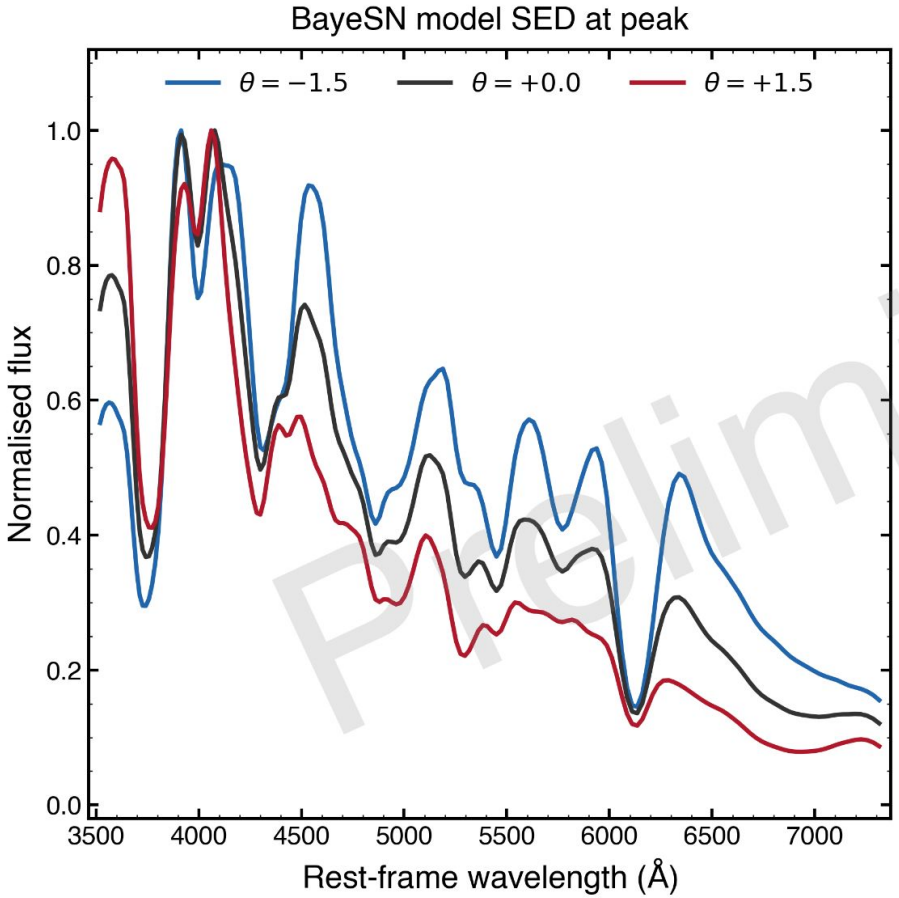
- *Something* is causing an intrinsic difference between SNe Ia in different environments
- This seems to be related to the i-band secondary maximum, need to investigate this further
- Maybe links to progenitor age
- Significant for astrophysics, also very significant for cosmology

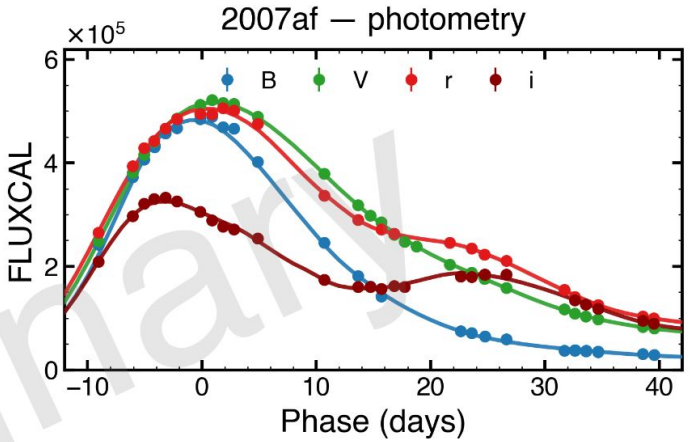


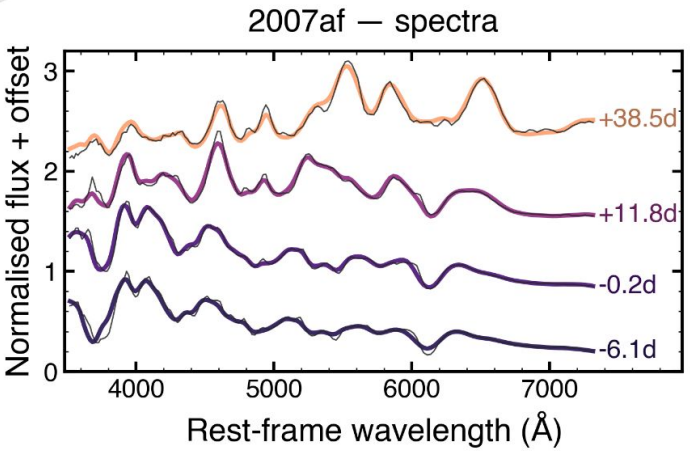
DESI Collaboration 2025

Next Steps

Next Step: Spectra



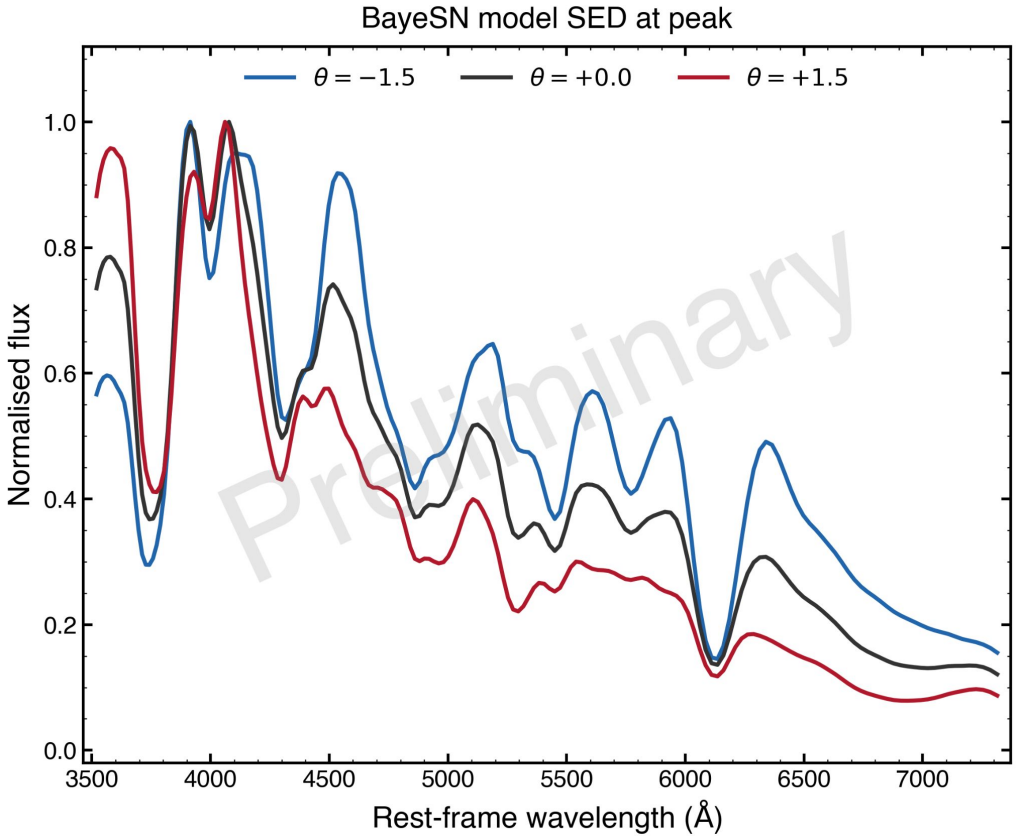




Preliminary

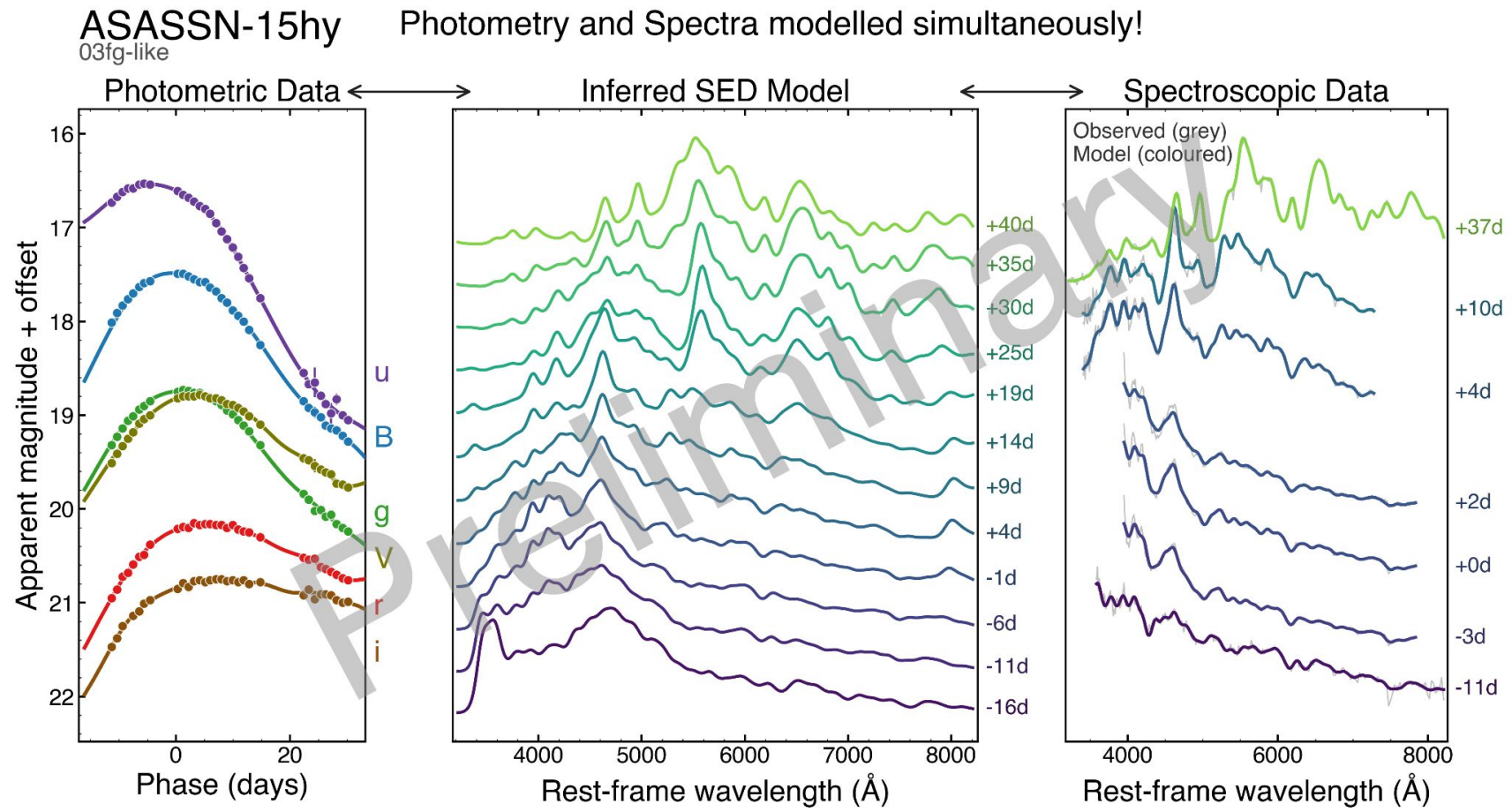
Next Step: Spectra

- Up until now, BayeSN has been applied purely to photometry; broad corrections to the Hsiao template
- Now, BayeSN can be applied to spectra and photometry simultaneously
- Should hopefully give reduced Hubble residual scatter
- Isolating intrinsic differences to specific spectral features will provide strong physical insight into what drives the mass step



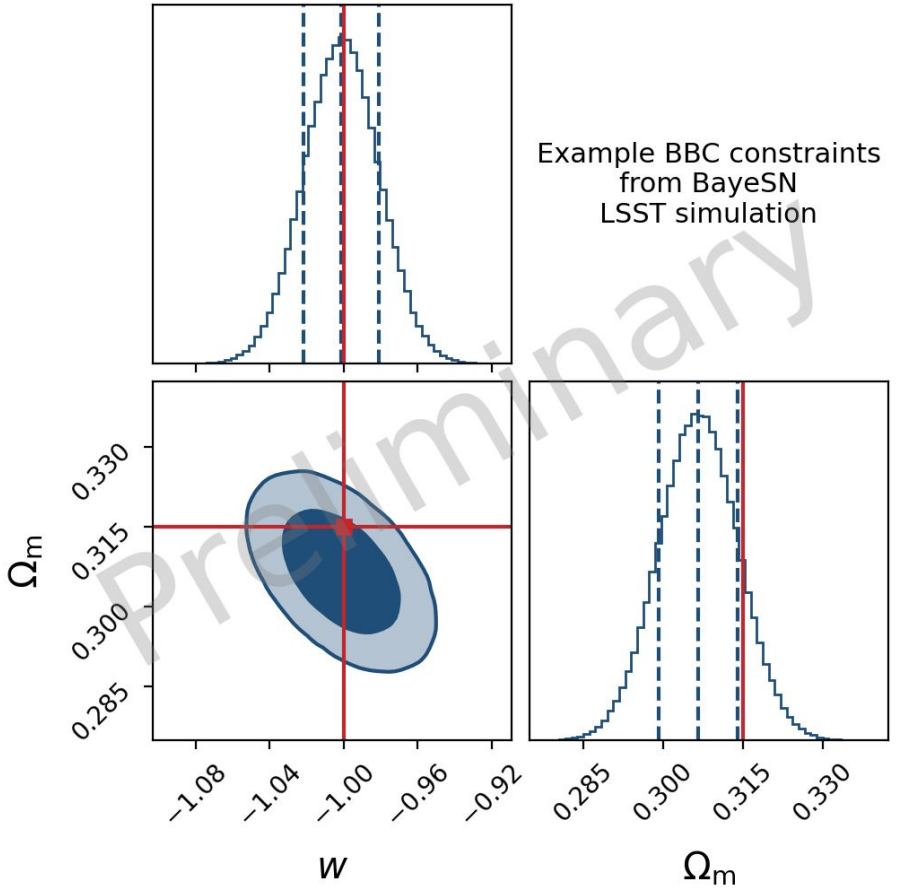
Grayling et al. in prep

Extending Beyond Type Ia Supernovae



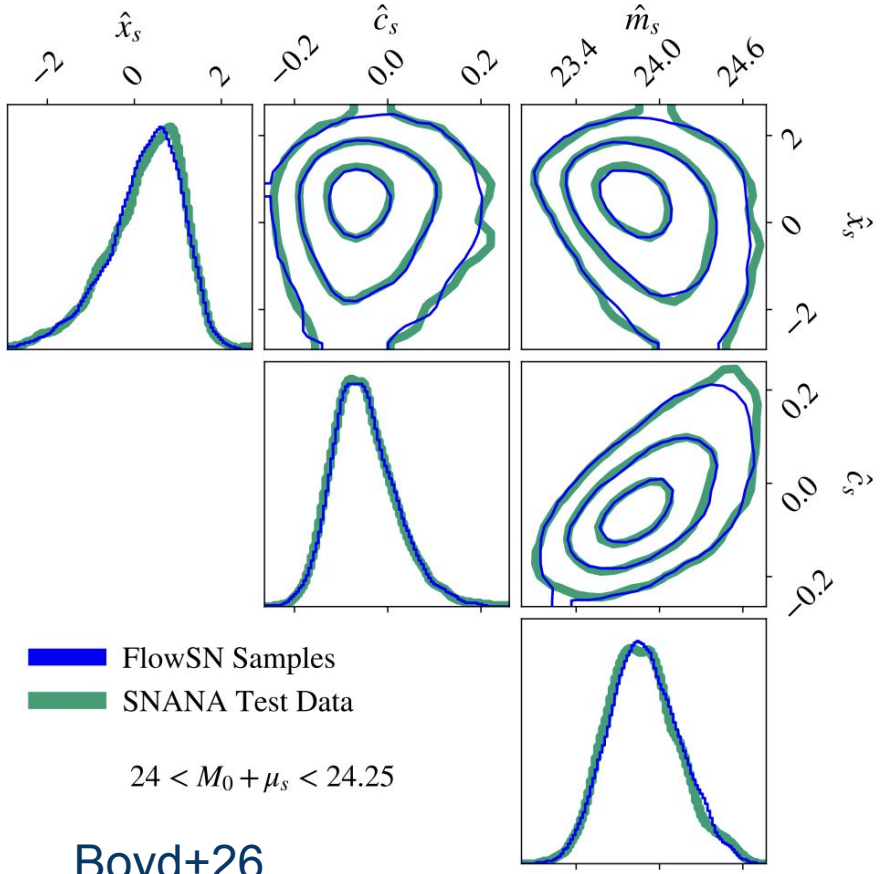
Cosmological Inference with BayeSN

1) Plugging BayeSN into existing SNANA cosmological pipeline
   - BBC approach using BayeSN simulations
2) Single-step cosmological inference
   - Infer cosmological parameters directly in the hierarchical model
   - Has key advantages, but requires modelling selection effects



Simulation-based Inference (SBI)

- Simply put…make a bunch of simulations, see which ones look like your data
- We are already good at simulating our survey observations
- Use neural networks to learn the complex relation between model parameters and data
- SBI relies on the ability to quickly generate lots of simulated datasets

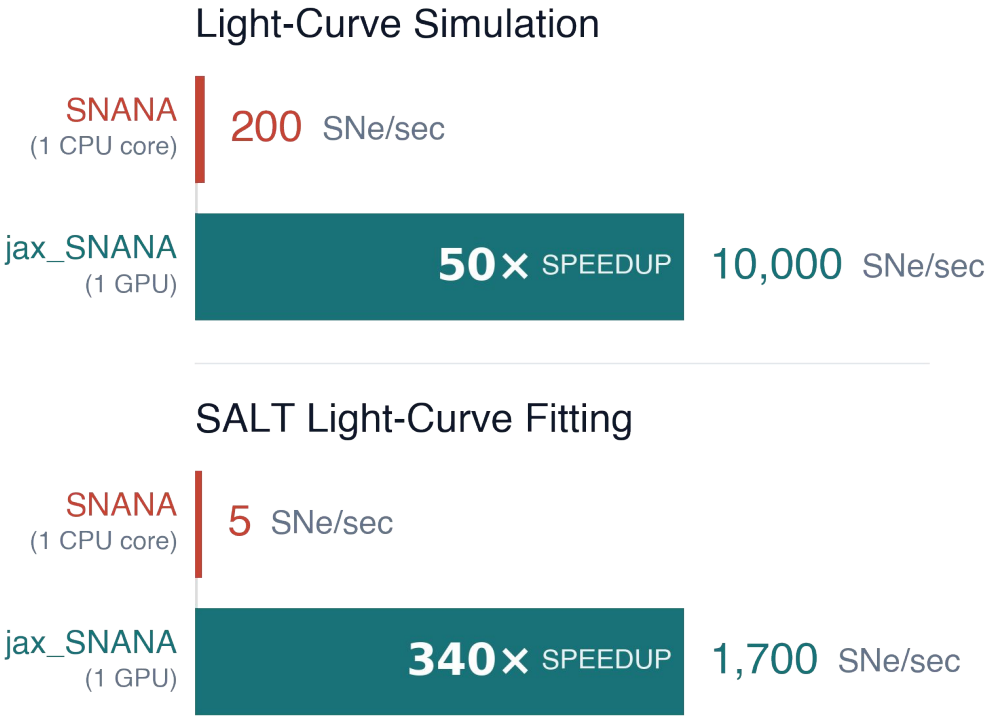


Boyd+26

jax_SNANA

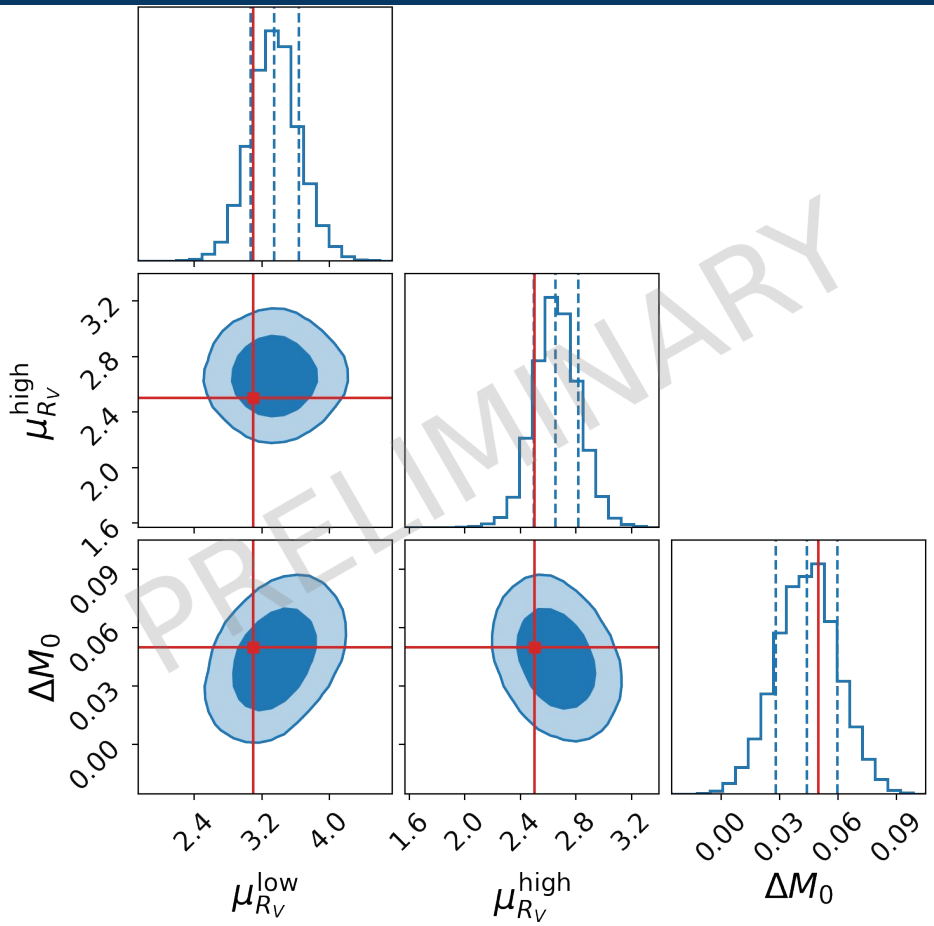
- SNANA in Python using jax, allowing it to run on a GPU
  - Big speedup, public release soon
  - Will be pip installable, no need to worry about your Fortran compiler!
- Includes SALT, BayeSN and a variety of core-collapse templates, being extended to include physically-motivated models using redback





Selection Effects in BayeSN

- jax_SNANA enables SBI with BayeSN, including selection effects
- We can accurately constrain R_V distributions and intrinsic mass step
- Will apply this to DES5YR, look at dust evolution with redshift
- Simple to extend the model to include cosmological parameters



Conclusions

- Supernova cosmology is at an exciting time, but limited by unknown astrophysics
- We can use BayeSN to empirically probe properties of the population of type Ia supernovae
- New G26 model out now, 12% reduced scatter on DES5YR
- Using jax_SNANA, we will infer these properties jointly with cosmological parameters using SBI

